



Oliver Nicholson

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Profile

Undergraduate Mechanical Engineer dedicated to advancing propulsion technologies within the aerospace industry. Combines strong foundations in thermofluids and heat transfer with practical experience in machining, CAD, and experimental testing to deliver innovative, high performance engineering solutions.

Education

Bachelor of Engineering (Mechanical) (Honours) 2022 - Present
Macquarie University

- Key coursework: Heat Transfer, Fluid Mechanics, Manufacturing Procedures, Materials and Systems Engineering
- Actively fostering student culture and advancing industry focused education initiatives to strengthen the future of the space sector
- Successfully advocated for new projects and initiatives within the Faculty of Science and Engineering

Employment

Engineering Research Intern Dec 2025 - Jan 2026
Macquarie University

- Contributed to engineering research in thermofluid systems, supporting design, modelling, and analysis of propulsion related concepts
- Assisted with iterative optimisation and experimental evaluation to improve system performance
- Collaborated with multidisciplinary teams, gaining hands on experience in technical problem solving and research driven engineering workflows

Coach and Session Lead Feb 2024 - Present
Grasshopper Soccer

- Coached and mentored young athletes at Grasshopper Soccer, delivering structured training sessions focused on skill development, coordination, and teamwork
- Managed groups in a dynamic environment, demonstrating strong leadership, communication, and adaptability
- Contributed to session planning and program delivery to ensure engaging and effective training outcomes

Funeral Director Jan 2022 - 2024
Gregory and Carr | Walker Group

- Assisted in the coordination and delivery of funeral services, supporting preparation, logistics, and client facing responsibilities
- Maintained a high level of professionalism and attention to detail in a sensitive environment
- Worked as part of a team to ensure services were conducted smoothly and respectfully, strengthening communication and organisational skills

Extracurricular Projects & Achievements

Manufacturing Lead, Macquarie Aerospace Rover Society (MARS) Jun 2024 - Present

Member of MARS, contributing to the design and development of rover and aerospace systems. Took on the role of lead machinist, supporting fabrication, prototyping, and assembly of components. Assisted with team coordination and workshops.

Skills

- Onshape
- Creo
- SolidWorks
- Fusion360
- Ansys
- MATLAB
- Python (NumPy, SciPy, Matplotlib)
- Thermofluid Systems Analysis
- Heat Transfer Modelling
- Regenerative Cooling Design
- Fluid Mechanics Application
- Pressure Drop Analysis
- Thermal Stress Analysis
- Computational Fluid Dynamics (CFD)
- Finite Element Analysis (FEA)
- Iterative Numerical Optimisation
- Data Analysis & Validation
- Manual Machining
- Tolerance Control & Fits
- Additive Manufacturing (FDM & SLM)
- Design for Manufacture (DFM)
- GD&T (Geometric Dimensioning & Tolerancing)
- Engineering Drawings & Technical Documentation
- Systems Engineering Design
- Multidisciplinary Integration
- Engineering Problem Solving

References

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